Department of Computer Science and Information Technology

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Odd Semester

DATA STRUCTURES AND ALGORITHMS-3

(25CS2103E)

< Movie Recommendation >

<P. Praveena> <2520090057>

<D. Varun Teja Reddy> <2520090221>

<P. Akhil> <2520090155>

COURSE INSTRUCTOR

Dr. Dyuti Banerjee
Professor
Department of Computer Science and Information Technology

Movie Recommendation

1. Introduction

Movie Recommendation is a Python Flask-based website that helps users discover movies according to their preferred language and genre. It supports Telugu, Hindi, Tamil, Kannada and Malayalam films. Users can explore movie posters, search titles, select a language and a genre, and view suitable recommendations with ratings and short descriptions.

The project demonstrates Data Structures and Algorithms through Python lists and dictionaries for storing movie records, Linear Search for finding matches, and Bubble Sort for arranging matching movies by rating. Each movie is assigned one main genre so that filtering is simple and consistent.

2. Problem Statement

Finding a suitable movie can be time-consuming when viewers must browse many titles across languages and categories. A user may want a Telugu romance film or a Malayalam feel-good movie but may not know which titles to choose. The Movie Recommendation website reduces this effort by providing a clear language-and-genre search and an ordered list of relevant movies.

3. Objectives of the Project

• Organise movies from Telugu, Hindi, Tamil, Kannada and Malayalam in one website.

• Store each movie as a Python dictionary in a list with title, language, one genre, year, rating, poster and description.

• Use Linear Search to find movies matching user-selected filters or a title query.

• Use Bubble Sort to rank matching movies from highest to lowest rating.

• Display movie cards, correct posters where available, movie details and related recommendations.

• Demonstrate practical DSA concepts in a simple, understandable Python application.

4. Application of Data Structures and Algorithms

A. Python Lists

A list stores the collection of movie records. The search algorithm checks the records one by one.

B. Python Dictionaries

Each movie is represented by a dictionary containing its name, original language, single main genre, release year, rating, poster reference and short description.

C. Linear Search

Linear Search scans each movie and compares its language, genre or title with the user input. For n movie records, the worst-case time complexity is O(n).

D. Bubble Sort

Bubble Sort compares neighbouring matched movies and swaps them when they are in the wrong rating order. It arranges higher-rated movies first. Its worst-case time complexity is $O(k^2)$ for k matched movies.

5. Practical Case Study Examples

Case Study 1: Telugu Romance Recommendations

A user selects Telugu and Romance. The system searches for matching titles such as Sita Ramam, Hi Nanna, Geetha Govindam, Darling and Mr. Perfect, then sorts the available results by rating.

Case Study 2: Malayalam Feel-Good Movies

A user selects Malayalam and Feel-Good. The website can recommend matching titles such as Home, Ustad Hotel, Charlie and Bangalore Days, depending on the stored dataset.

Case Study 3: Movie Name Search

A user searches for Magadheera. The system compares the search text with movie titles and displays the matching film card and its details.

Case Study 4: Single Genre Classification

The movie RRR is assigned Action only, rather than being repeated under Action and Drama. This keeps the category filter predictable and avoids duplicate entries.

6. Expected Results and Benefits

- A single interface for discovering movies in five Indian languages.
- Recommendations that match the selected language and one main genre.
- Higher-rated matching movies displayed first.
- Searchable movie names and movie-detail pages.
- A simple practical demonstration of Linear Search and Bubble Sort.

7. Conclusion

Movie Recommendation demonstrates how basic Data Structures and Algorithms can support a useful web application. Lists and dictionaries organise movie data, Linear Search identifies relevant movies, and Bubble Sort ranks the results. The Python Flask website offers a straightforward way to discover films by language and genre.

8. Future Scope

The website could later include more movies, user ratings, watchlists, improved search, verified poster integrations and advanced recommendation techniques. These are future enhancements, not part of the present DSA-based recommendation logic.

ALGORITHM / PSEUDOCODE

1. Overall Recommendation Algorithm

```
ALGORITHM Movie_Recommendation
BEGIN
  LOAD movie list
  DISPLAY languages and genres
  INPUT selectedLanguage, selectedGenre
  matches ← Linear_Search(movies, selectedLanguage, selectedGenre)
  ranked ← Bubble_Sort(matches)
  DISPLAY ranked movie cards
END
```

2. Linear Search Algorithm

```
ALGORITHM Linear_Search(movies, language, genre)
BEGIN
  matches ← empty list
  FOR each movie IN movies DO
    IF (language = "All" OR movie.language = language) AND
       (genre = "All" OR movie.genre = genre) THEN
      APPEND movie TO matches
    END IF
  END FOR
  RETURN matches
END
```

3. Bubble Sort Algorithm

```
ALGORITHM Bubble_Sort(matches)
BEGIN
  FOR i ← 0 TO length(matches) - 2 DO
    FOR j ← 0 TO length(matches) - i - 2 DO
      IF matches[j].rating < matches[j+1].rating THEN
        SWAP matches[j], matches[j+1]
      END IF
    END FOR
  END FOR
  RETURN matches
END
```

4. Movie Title Search

```
ALGORITHM Search_Title(movies, query)
BEGIN
  results ← empty list
  FOR each movie IN movies DO
    IF lowercase(query) is contained in lowercase(movie.name) THEN
      APPEND movie TO results
    END IF
  END FOR
  RETURN results
END
```

5. Complexity and Algorithm Analysis

Operation	Approach	Worst-case time
Language and genre filtering	Linear Search	O(n)
Movie title search	Linear Search	O(n)
Ranking matched movies	Bubble Sort	$O(k^2)$

Here n is the number of movie records and k is the number of matching movies. The sorting cost depends on the number of matches, not necessarily the entire dataset.

6. Conclusion

The pseudocode shows the main steps used in the Movie Recommendation project: accept the user's preferences, scan stored movie records, sort matching films by rating and display the results. These steps demonstrate the direct use of DSA in a Python website.

RESULTS

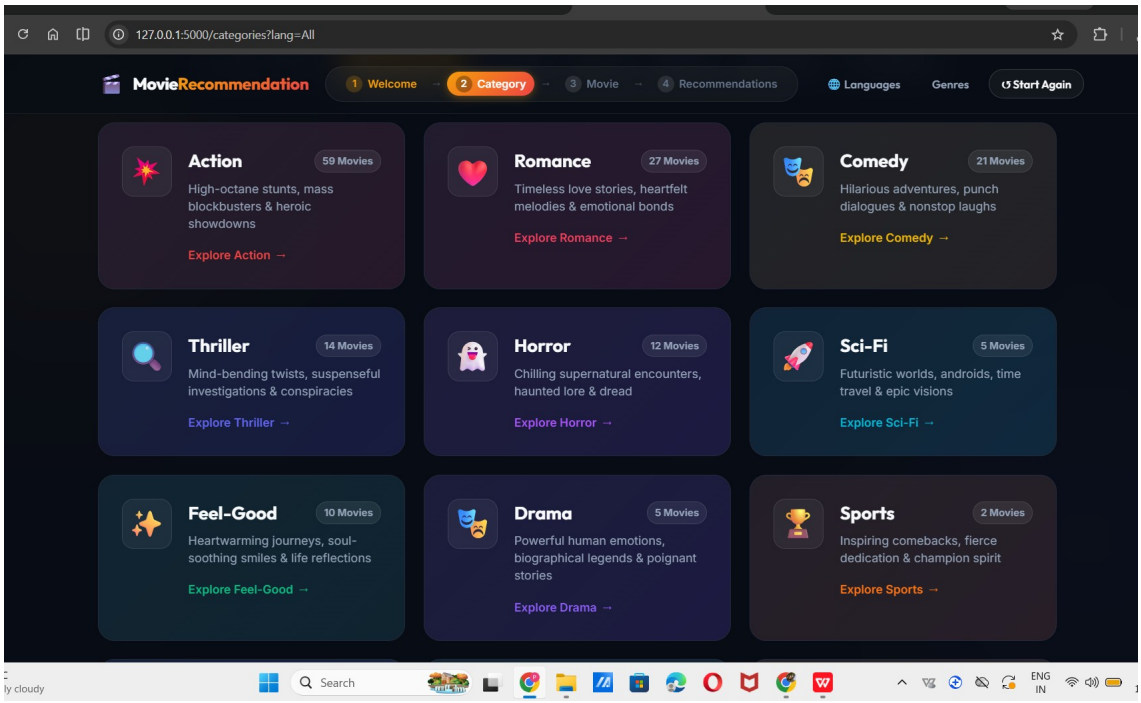
Screenshots

The Movie Recommendation website contains a homepage, five language categories, genre filters, a movie search feature, recommendation results and movie details. The following screenshots should be inserted from the actual running website:

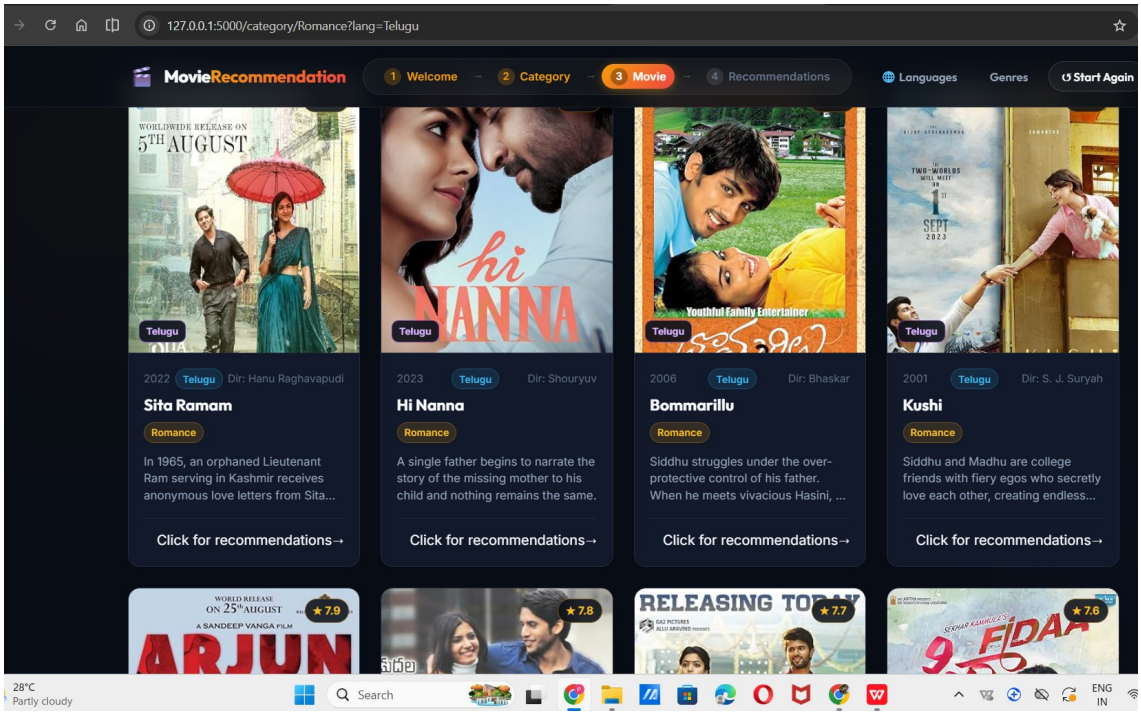
Screenshot 1: Homepage – Movie Recommendation



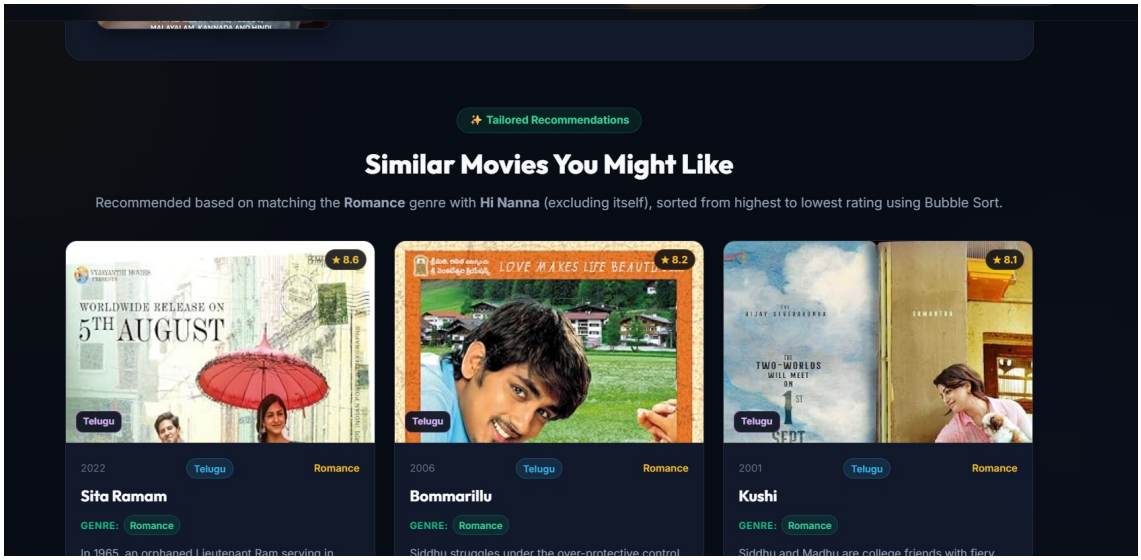
Screenshot 2: Movie Genre Categories



Screenshot 3: Telugu Romance Movie Listings



Screenshot 4: Similar Movie Recommendations



Screenshot 5: Additional Genre Categories

